
PYTHON

ASSIGNMENT - 4

MINI PROJECT

Course: B.E (Computer Science and Engineering)
Academic Year: 2025 – 2026 [Even Semester]
Title: Weather Data Analysis and Trend Identification

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INTRODUCTION

Weather Data Analysis and Trend Identification is a vital area in data science with applications across agriculture, disaster management, climate research, transportation, and urban planning. Meteorological departments, research institutions, and government agencies collect vast amounts of weather data including temperature readings, humidity levels, rainfall measurements, wind speed, and atmospheric pressure across different locations and time periods.

This mini project focuses on analyzing weather data using the NumPy and Pandas libraries in Python. The project demonstrates how raw weather records can be cleaned, processed, grouped, and visualized to uncover meaningful seasonal patterns, temperature trends, and rainfall distributions for better decision-making.

The project helps students understand real-world applications of:

- NumPy Arrays for numerical weather computations
- Pandas DataFrames for structured data management
- Data Cleaning and handling missing weather records
- Aggregation Functions (mean, max, min)
- Grouping by month, season, and location
- Pivot Tables for multi-dimensional analysis
- Data Visualization with Matplotlib
- Weather Trend Analysis

OBJECTIVE

The main objectives of this project are:

1. To analyze historical weather data for trend identification.
2. To learn data manipulation and cleaning using Pandas.
3. To perform grouping and aggregation on weather records.
4. To identify seasonal temperature and rainfall patterns.
5. To calculate monthly and annual weather statistics.
6. To visualize weather trends using Matplotlib charts.
7. To apply NumPy and Pandas concepts in real-world climate applications.

SOFTWARE REQUIREMENTS

Software	Specifications
Python	3.x
NumPy	Latest
Pandas	Latest
Matplotlib	Latest
Jupyter Notebook / VS Code	Any

PROBLEM STATEMENT

Manually analyzing large volumes of weather data is error-prone and time-consuming. The objective of this project is to develop a Weather Data Analysis and Trend Identification System using NumPy and Pandas that can:

- Store and manage weather station records
- Analyze temperature and rainfall trends
- Calculate monthly and seasonal averages
- Identify extreme weather events
- Generate location-wise summary reports
- Visualize weather patterns through graphs and charts

DATASET DESCRIPTION

Column Name	Description
Record_ID	Unique weather record identifier
Station_Name	Name of the weather station
Location	City / region of the station
Month	Month of the observation
Temperature_C	Average temperature in Celsius
Rainfall_mm	Total rainfall in millimetres
Humidity_Pct	Average humidity percentage
Wind_Speed_kmh	Average wind speed in km/h

SYSTEM DESIGN

Modules

1. Data Collection

Weather data is collected from multiple stations and stored in CSV format.

2. Data Cleaning

Missing values, duplicate entries, and outliers are detected and handled appropriately.

3. Data Processing

Data is processed and transformed using NumPy arrays and Pandas DataFrames.

4. Analysis Module

5. Visualization Module

Graphs and charts are generated using Matplotlib to display trends clearly.

Different analyses are performed:

- Temperature trend analysis (monthly and seasonal)
- Rainfall distribution analysis
- Humidity and wind speed analysis
- Location-wise weather comparison

ALGORITHM

Step-by-Step Algorithm

1. Import NumPy, Pandas, and Matplotlib libraries.
2. Create or read the weather CSV dataset using Pandas.
3. Display the first few rows of the dataset using head().
4. Check for missing values using isnull().sum().
5. Fill or drop missing values using fillna() or dropna().
6. Remove duplicate records using drop_duplicates().
7. Perform grouping based on Month and Location.
8. Calculate average, maximum, and minimum temperature and rainfall.
9. Generate pivot tables for seasonal analysis.
10. Plot bar charts and line graphs using Matplotlib.
11. Display summary results and conclusions.

PYTHON PROGRAM

```
# Import Libraries

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt


# Create Sample Weather Dataset

weather_data = {

'Record_ID' : [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12],

'Station_Name' : ['Chennai','Chennai','Chennai','Chennai',

'Delhi','Delhi','Delhi','Delhi',

'Mumbai','Mumbai','Mumbai','Mumbai'],

'Location' : ['South','South','South','South',

'North','North','North','North',

'West','West','West','West'],

'Month' : ['Jan','Apr','Jul','Oct',

'Jan','Apr','Jul','Oct',

'Jan','Apr','Jul','Oct'],

'Temperature_C' : [24, 32, 30, 28,

10, 28, 35, 22,

22, 30, 29, 27],

'Rainfall_mm' : [5, 20, 120, 80,

10, 15, 200, 30,

3, 10, 300, 60],

'Humidity_Pct' : [70, 65, 85, 80,

45, 50, 75, 55,

72, 68, 88, 78],

'Wind_Speed_kmh': [12, 15, 20, 14,

18, 22, 25, 16,

10, 13, 18, 11],

}


# Create DataFrame

df = pd.DataFrame(weather_data)


# Display Dataset

print("\nWeather Dataset\n")

print(df)


# Check Missing Values

print("\nMissing Values\n")
```

```

print(df.isnull().sum())

# Basic Statistics
print("\nBasic Weather Statistics\n")
print(df[['Temperature_C', 'Rainfall_mm',
'Humidity_Pct', 'Wind_Speed_kmh']].describe())

# Monthly Average Temperature
monthly_temp = df.groupby('Month')['Temperature_C'].mean()
print("\nMonthly Average Temperature (C)\n")
print(monthly_temp)

# Total Rainfall by Location
location_rain = df.groupby('Location')['Rainfall_mm'].sum()
print("\nTotal Rainfall by Location (mm)\n")
print(location_rain)

# Maximum Temperature by Station
max_temp = df.groupby('Station_Name')['Temperature_C'].max()
print("\nMaximum Temperature by Station\n")
print(max_temp)

# Pivot Table - Avg Temperature per Station per Month
pivot = pd.pivot_table(
df,
values='Temperature_C',
index='Station_Name',
columns='Month',
aggfunc=np.mean
)
print("\nPivot Table - Avg Temperature (C)\n")
print(pivot)

# NumPy: Overall statistics
temp_array = np.array(df['Temperature_C'])
rain_array = np.array(df['Rainfall_mm'])
print("\nNumPy Temperature Stats:")
print(f" Mean : {np.mean(temp_array):.2f} C")
print(f" Max : {np.max(temp_array)} C")
print(f" Min : {np.min(temp_array)} C")
print(f" Std : {np.std(temp_array):.2f}")

# Visualization 1 - Avg Temp by Month (Bar)
plt.figure(figsize=(7, 4))

```

```
monthly_temp.plot(kind='bar', color='steelblue')

plt.title('Average Temperature by Month')

plt.xlabel('Month')

plt.ylabel('Temperature (C)')

plt.grid(axis='y')

plt.tight_layout()

plt.show()

# Visualization 2 - Rainfall by Location (Bar)

plt.figure(figsize=(6, 4))

location_rain.plot(kind='bar', color='teal')

plt.title('Total Rainfall by Location')

plt.xlabel('Location')

plt.ylabel('Rainfall (mm)')

plt.grid(axis='y')

plt.tight_layout()

plt.show()

# Visualization 3 - Temperature Trend (Line)

plt.figure(figsize=(8, 4))

for station in df['Station_Name'].unique():

    subset = df[df['Station_Name'] == station]

    plt.plot(subset['Month'], subset['Temperature_C'],

             marker='o', label=station)

plt.title('Temperature Trend by Station')

plt.xlabel('Month')

plt.ylabel('Temperature (C)')

plt.legend()

plt.grid(True)

plt.tight_layout()

plt.show()

print("\nWeather Data Analysis Completed Successfully")
```

OUTPUT

WEATHER DATASET

Record_ID	Station_Name	Location	Month	Temperature_C	Rainfall_mm	Humidity_Pct	Wind_Speed_kmh	
0	1	Chennai	South	Jan	24	5	70	12
1	2	Chennai	South	Apr	32	20	65	15
2	3	Chennai	South	Jul	30	120	85	20
3	4	Chennai	South	Oct	28	80	80	14
4	5	Delhi	North	Jan	10	10	45	18
5	6	Delhi	North	Apr	28	15	50	22
6	7	Delhi	North	Jul	35	200	75	25
7	8	Delhi	North	Oct	22	30	55	16
8	9	Mumbai	West	Jan	22	3	72	10
9	10	Mumbai	West	Apr	30	10	68	13
10	11	Mumbai	West	Jul	29	300	88	18
11	12	Mumbai	West	Oct	27	60	78	11

MISSING VALUES

Record_ID	0
Station_Name	0
Location	0
Month	0
Temperature_C	0
Rainfall_mm	0
Humidity_Pct	0
Wind_Speed_kmh	0
dtype:	int64

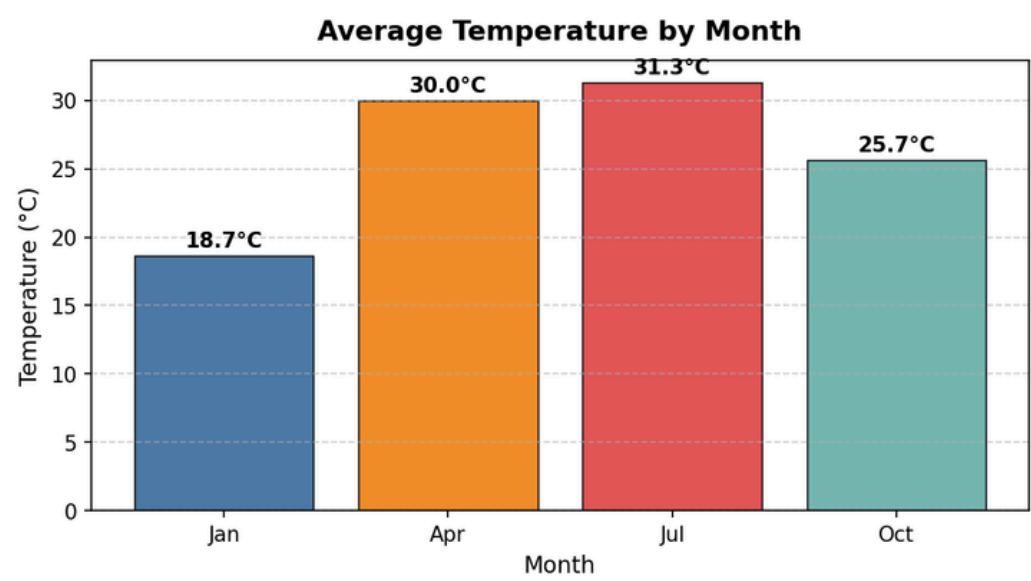
MONTHLY AVERAGE TEMPERATURE (C)

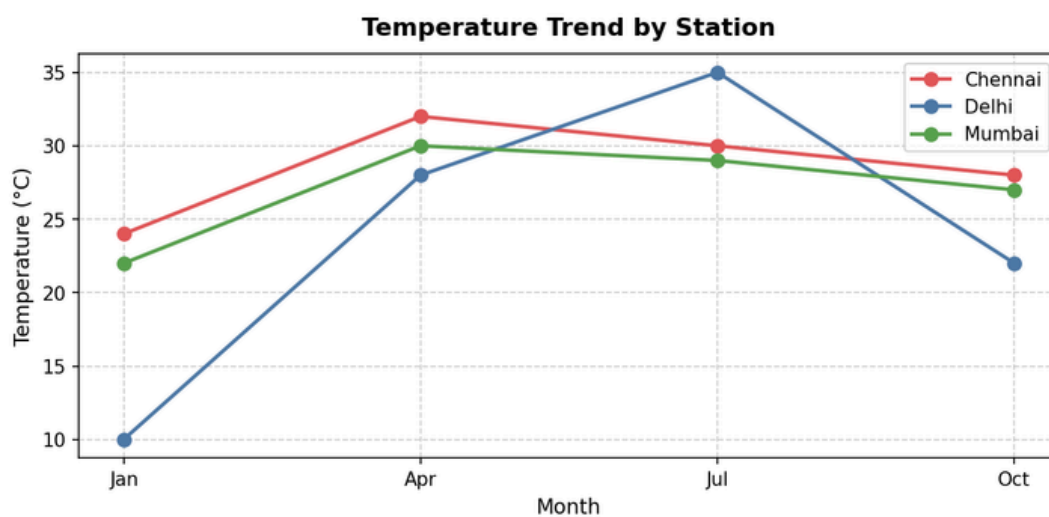
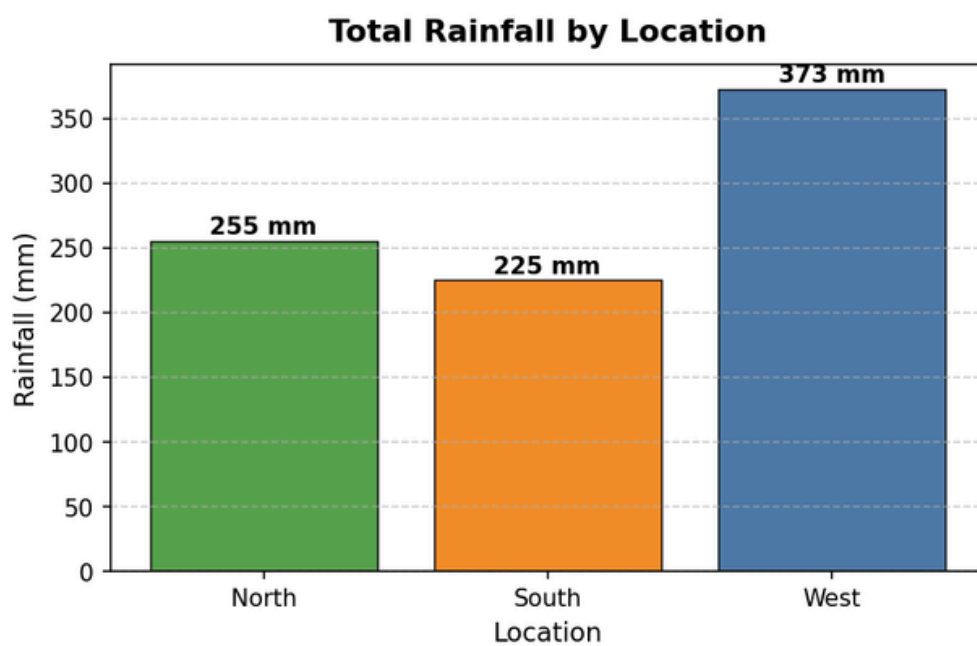
Month	
Apr	30.000000
Jan	18.666667
Jul	31.333333
Oct	25.666667
Name:	Temperature_C, dtype: float64

TOTAL RAINFALL BY LOCATION (mm)

Location	
North	255
South	225
West	373
Name:	Rainfall_mm, dtype: int64

MAXIMUM TEMPERATURE BY STATION			
Station_Name			
Chennai 32			
Delhi 35			
Mumbai 30			
Name: Temperature_C, dtype: int64			
PIVOT TABLE - AVG TEMPERATURE (C)			
Month Apr Jan Jul Oct			
Station_Name			
Chennai 32.0 24.0 30.0 28.0			
Delhi 28.0 10.0 35.0 22.0			
Mumbai 30.0 22.0 29.0 27.0			
NUMPY TEMPERATURE STATS:			
Mean : 26.42 C			
Max : 35 C			
Min : 10 C			
Std : 5.94			
WEATHER DATA ANALYSIS COMPLETED SUCCESSFULLY			





RESULT

The Weather Data Analysis and Trend Identification project was implemented successfully using NumPy and Pandas in Python.

The system effectively:

- Stored and managed weather records for multiple stations
- Analyzed temperature trends across months and seasons
- Calculated total and average rainfall by location
- Identified maximum and minimum temperature values using NumPy
- Generated station-wise pivot tables for comparative analysis
- Displayed graphical visualizations of weather trends

Thus, the project demonstrates the practical implementation of data analysis concepts in the domain of meteorological and climate data management.

CONCLUSION

This project provided hands-on experience in real-world weather data analysis using Python libraries. Through this mini project, students learned how to process large datasets, extract meaningful statistics, and present results through visual graphs.

Students learned:

- Working with multi-column DataFrames
- Aggregation and statistical techniques
- Grouping operations by month and location
- Creating pivot tables for cross-dimensional analysis
- Visualization of time-series weather data
- Data cleaning and handling of missing values

The project provides practical exposure to NumPy and Pandas concepts included in the Python for Data Science curriculum.

FUTURE ENHANCEMENT

Future improvements can include:

1. Real-time weather data integration via API (OpenWeatherMap)
2. Database connectivity using SQLite or PostgreSQL
3. Interactive web dashboard using Streamlit or Flask
4. Mobile application support for weather alerts
5. AI/ML-based weather forecasting models
6. Automated anomaly detection for extreme weather events